AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 35 · WORKBOOK

Improve and Challenge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected repair record, with a testing extension.

Locate the cause, test neighbors, reject a bad patch and write the change log.

Keep one habit: Retain original and revised evidence, concrete expected results, version and unresolved limits.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Test Diverse Users. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Test Diverse Users

Project Problem and Success Test

Evidence and Sources

W1 · Locate the strict-comparison cause

> Explain why at most includes equality.

Your evidence record

- Main boundary contract: validated duration at most limit means minutes<=limit. At limit10,9/10/11→True/True/False. A schema-invalid row must never reach selection; zero is valid. Activity score2fun+learning; equal eligible scores use ascending name. Separate page picture: pages<=100, highest interest, ID tie; A100/5,B90/3,C101/5. Old<100selectsB;repaired<=100selectsA;bad<=101admitsC.

- Why does minutes < limit reject a valid exactly-at-limit activity?

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Test neighboring durations

> Keep below,at,above expectations visible.

Your evidence record

- Main boundary contract: validated duration at most limit means minutes<=limit. At limit10,9/10/11→True/True/False. A schema-invalid row must never reach selection; zero is valid. Activity score2fun+learning; equal eligible scores use ascending name. Separate page picture: pages<=100, highest interest, ID tie; A100/5,B90/3,C101/5. Old<100selectsB;repaired<=100selectsA;bad<=101admitsC.

- For limit 15, give expected eligibility at 14, 15 and 16.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Reject a limit-plus-one patch

> Supply the counterexample before running.

Your evidence record

- Main boundary contract: validated duration at most limit means minutes<=limit. At limit10,9/10/11→True/True/False. A schema-invalid row must never reach selection; zero is valid. Activity score2fun+learning; equal eligible scores use ascending name. Separate page picture: pages<=100, highest interest, ID tie; A100/5,B90/3,C101/5. Old<100selectsB;repaired<=100selectsA;bad<=101admitsC.

- A proposed fix accepts minutes <= limit + 1. Give a counterexample at limit 15.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Write a truthful change log

> Retain the bug, precise change, regression evidence and remaining limits.

Your evidence record

- Main boundary contract: validated duration at most limit means minutes<=limit. At limit10,9/10/11→True/True/False. A schema-invalid row must never reach selection; zero is valid. Activity score2fun+learning; equal eligible scores use ascending name. Separate page picture: pages<=100, highest interest, ID tie; A100/5,B90/3,C101/5. Old<100selectsB;repaired<=100selectsA;bad<=101admitsC.

- Record the bug, change, **evidence** and one remaining limit without claiming all bugs are gone.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Repair a new boundary independently

> Keep the first attempt before hints.

Your evidence record

- Use exact integer duration validation0..60 before the predicate. The independent12-minute task is separate from the worked10- and15-minute practices. Supply actual tie and invalid-data records plus expected reports; a label such as tie test is not a reproducible case. Preserve whether you wrote a prediction, made a paper trace or actually ran code.

- Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Repair independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Publish the cause, one change, fixed test set, **before/after evidence**, and remaining limitation.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Test neighbors for Improving without hiding failures. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A repair raises success from 7/10 to 9/10 on the same validation set.

What belongs in the improvement claim?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 35 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

My reasoning and evidence

▼ Extension · Testing functions and interfaces — W1 Choose test scale

REFERENCE · SECTION 17

Extension · Testing functions and interfaces — W1 Choose test scale

> Retained extension task with its own teacher key and response.

Extension question and record

- Classify testing score(3,2) versus running raw rows through the whole chooser.

My reasoning and evidence

▼ **Extension · Testing functions and interfaces — W2 Calculate an oracle**

REFERENCE · SECTION 18

Extension · Testing functions and interfaces — W2 Calculate an oracle

Retained extension task with its own teacher key and response.

Extension question and record

- What expected score should a test use for fun 4 and learning 3?

My reasoning and evidence

▼ Extension · Testing functions and interfaces — W3 Repair a weak test

REFERENCE · SECTION 19

Extension · Testing functions and interfaces — W3 Repair a weak test

> Retained extension task with its own teacher key and response.

Extension question and record

- A test compares recommend(items,10) to itself. Replace it with a useful assertion.

> **My reasoning and evidence**

▼ Extension · Testing functions and interfaces — W4 Test independently

REFERENCE · SECTION 20

Extension · Testing functions and interfaces — W4 Test independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Design an integration **fixture**: A valid (minutes 10, fun 2, learning 4), B valid (10,3,2), C invalid (5,9,1), limit 10. State winner, counts and a reversed-order expectation.

My reasoning and evidence

▼ **Extension · Testing functions and interfaces — W5 Cover absence**

REFERENCE · SECTION 21

Extension · Testing functions and interfaces — W5 Cover absence

Retained extension task with its own teacher key and response.

Extension question and record

- What expected result should an empty-**input** integration case specify?

My reasoning and evidence

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional recommender excludes books at a page limit that is intended to include equality.

Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

AI ACADEMY / AI-2/35

Improve and Challenge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify cause

A wrongly excluded at equality, so old code selects B.

2. Focused repair

Replace <100 with <=100 without changing ranking or type checks.

3. Expected revised result

A becomes eligible and wins; C remains excluded.

4. Challenge repair

Rerun 99,100,101; invalid pages; tied scores; reversed row order.

CHANGE ONE THING

An alternative patch simply increases the limit to 101.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why are tests at both 100 and 101 needed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional recommender excludes books at a page limit that is intended to include equality.
- Which labelled step matches this detail? Rerun 99,100,101; invalid pages; tied scores; reversed row order.
- 1. Identify cause
- 2. Expected revised result
- 3. Challenge repair
- 4. Focused repair
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ____
>
> ____
>
> ____
>
> ____

▼ **Check after trying**

After your attempt, compare with the supported explanation: Challenge repair Rerun 99,100,101; invalid pages; tied scores; reversed row order.

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Changed-case question

- Changed-case practice: An alternative patch uses pages <= 101 instead of the required pages <= 100. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. C is wrongly admitted as well as A.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: C is wrongly admitted as well as A. Changing the maximum alters the contract instead of repairing its comparison.

Repair and keep evidence

Review Test Diverse Users (ai-2/34). Goal: Evaluate across varied cases. Short practice: Can these records establish performance for different user groups?. Return to this week and explain what you changed.

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

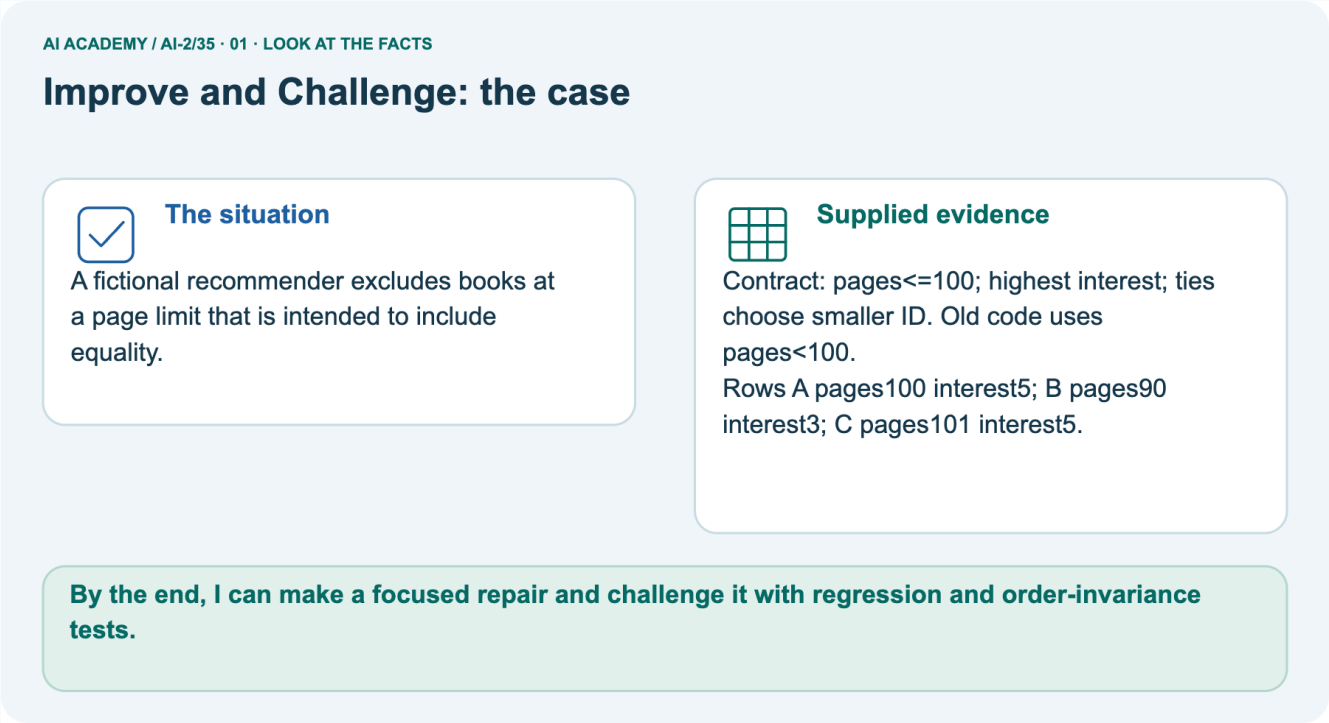
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

Choose the relevant picture rather than repeating every task.

01. Improve and Challenge: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

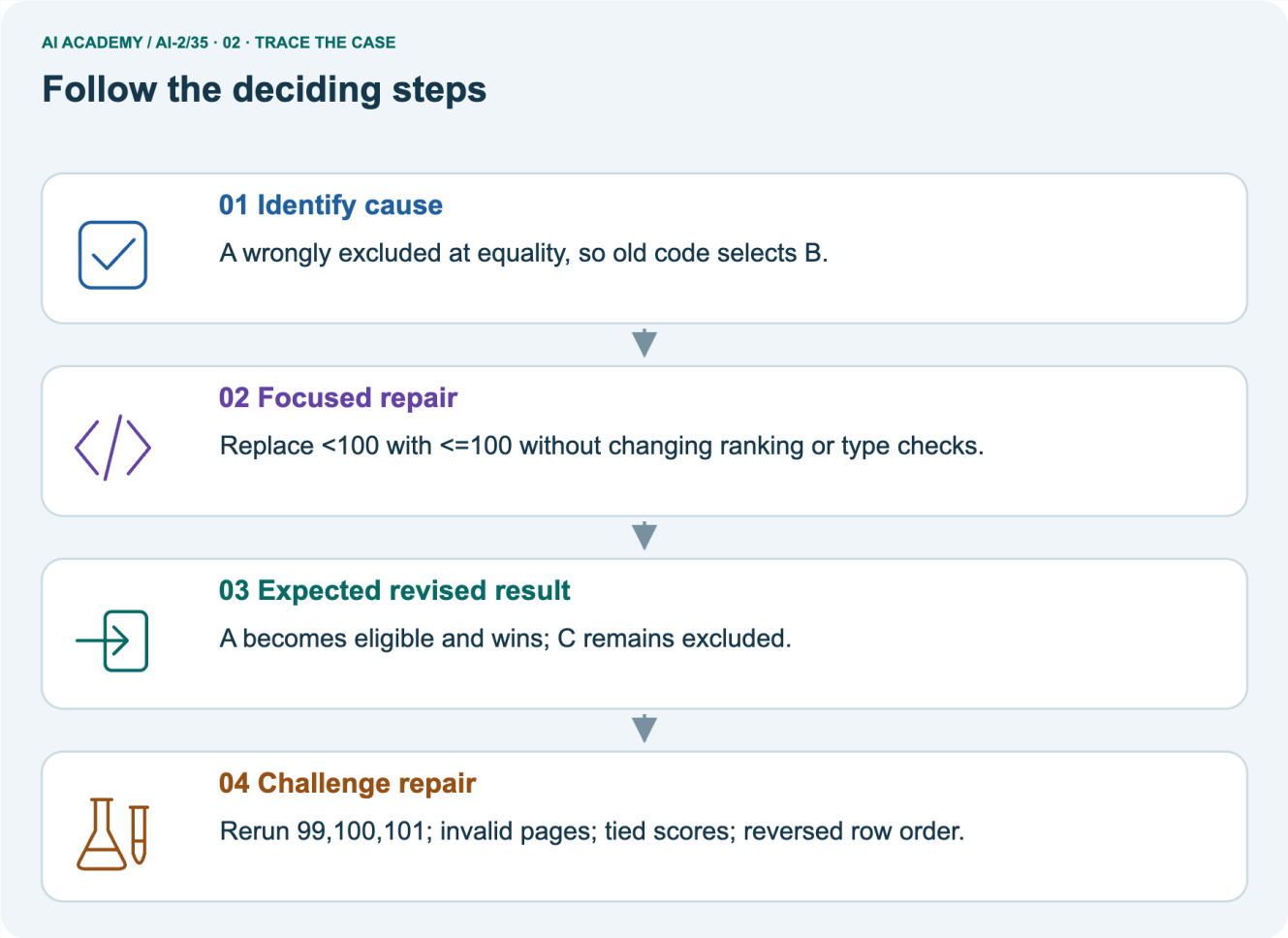
A fictional recommender excludes books at a page limit that is intended to include equality.
Supplied facts: Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



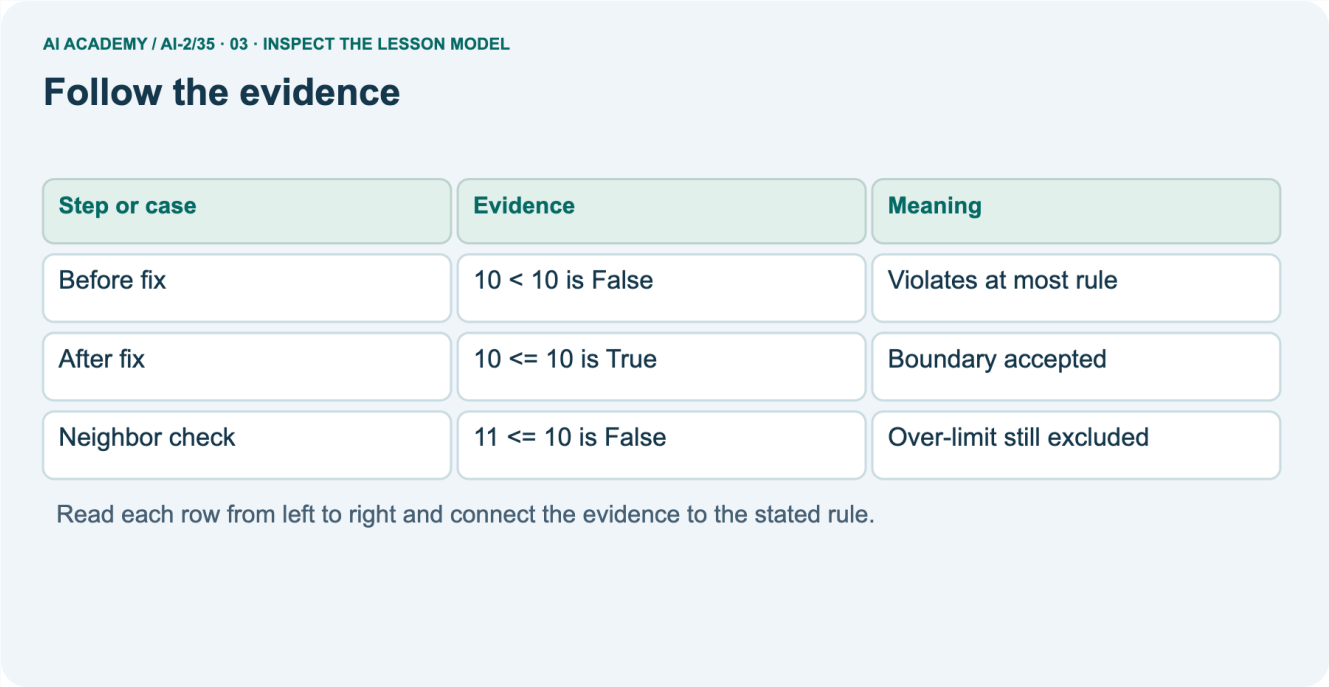
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify cause: A wrongly excluded at equality, so old code selects B. → Focused repair: Replace <100 with <=100 without changing ranking or type checks. → Expected revised result: A becomes eligible and wins; C remains excluded. → Challenge repair: Rerun 99,100,101; invalid pages; tied scores; reversed row order.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



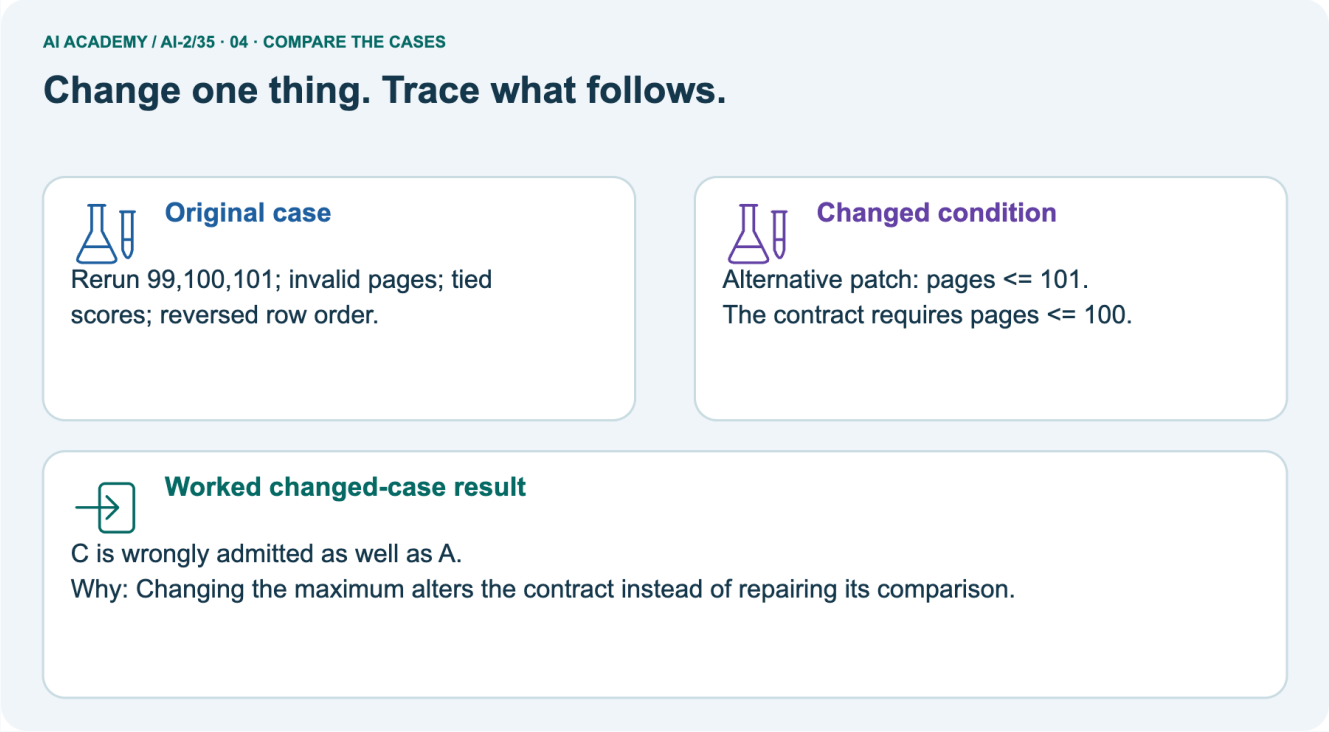
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Before fix | 10 < 10 is False | Violates at most rule. After fix | 10 <= 10 is True | Boundary accepted. Neighbor check | 11 <= 10 is False | Over-limit still excluded

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



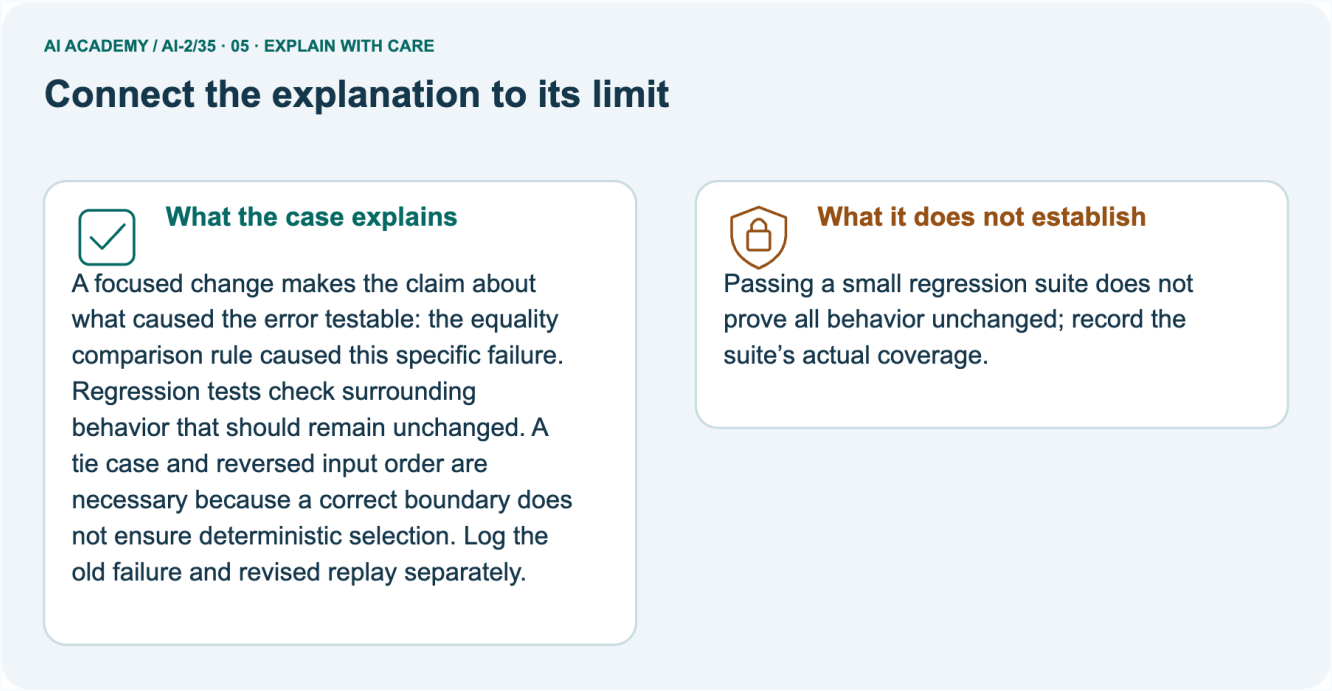
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Rerun 99,100,101; invalid pages; tied scores; reversed row order. Changed condition: An alternative patch uses pages <= 101 instead of the required pages <= 100. Expected result: C is wrongly admitted as well as A. Reason: Changing the maximum alters the contract instead of repairing its comparison.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

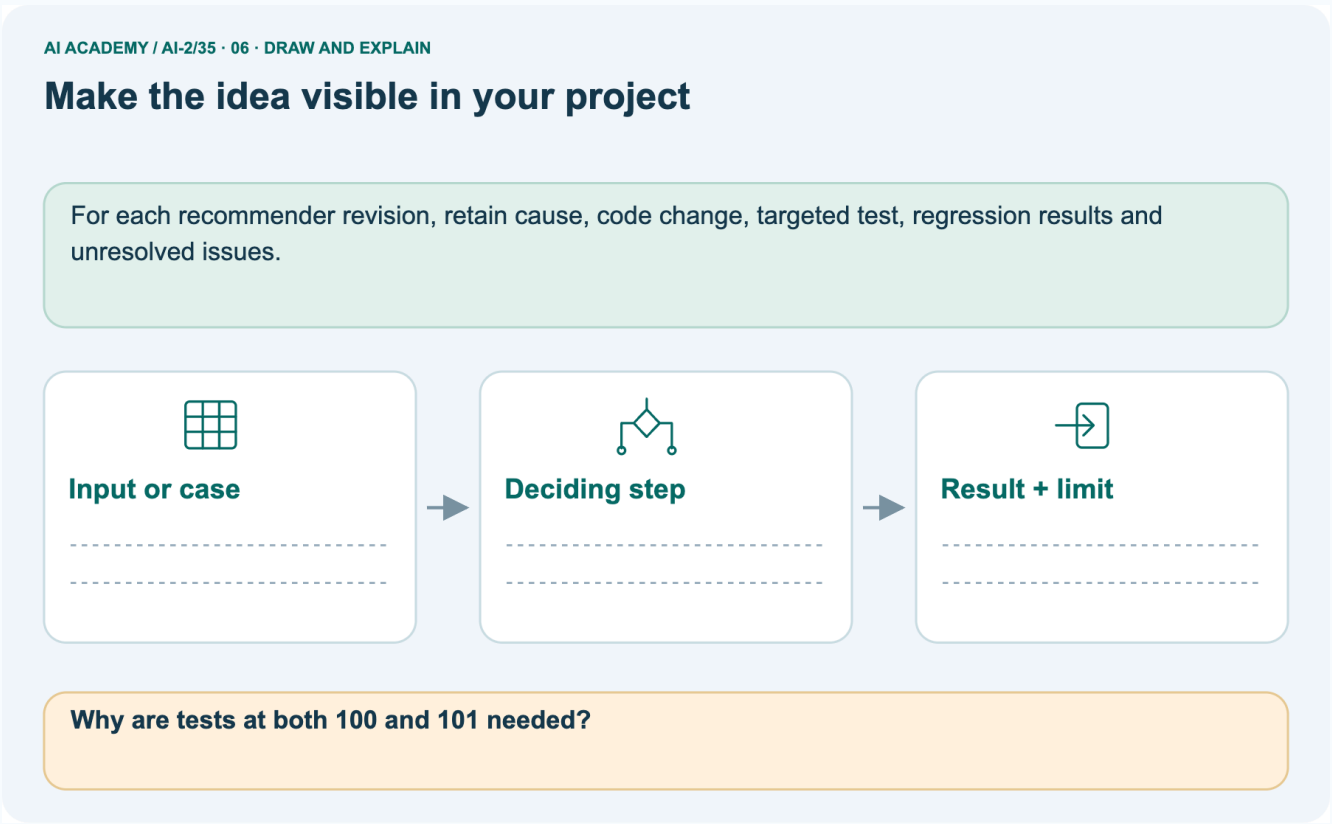
A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Limit: Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues. Draw your **input**, deciding step, result and limit. Supported question: Why are tests at both 100 and 101 needed?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Improve and Challenge: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.